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Report

Gym Management System

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Presentation link:
https://www.canva.com/design/DAGFtTCG8Ss/xcsWUxGjjEoCkUKtdsJvYA/edit?utm_content=DAGFtTCG8Ss&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

GitHub link: https://github.com/GhalaMansour/gym_system

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Introduction:

The Gym Management is a system designed to assist in the efficient management of a gym facility. This system allows gym administrators to perform various tasks such as adding new members, managing trainers, scheduling classes, and monitoring membership expiration.

Features:

- **Member Management:**
  - adding new members
  - viewing the member list
  - checking membership expiration.

- **Trainer Management:**
  - Add new trainer
  - View trainer list

- **Classes management :**
  - Add new class
  - View class schedule

explaining the code

Helper Functions for File-based Storage

- **save_data_to_file():** This function is used to save data to a file. It checks if the file already exists, and if so, it appends the new data to the existing file. If the file doesn't exist, it creates a new file with the provided data.
- **read_data_from_file():** This function reads the contents of a file and returns the data. If the file doesn't exist, it returns an empty string.

Function to Create an Invoice

- **create_invoice():** This function generates an invoice for a new member or an existing member with a new membership. It takes the membership duration and fee as input parameters,

generates a random invoice number, and the current date. It then creates an invoice data string and saves it to the "invoice_data.txt" file using the `save_data_to_file()` function.

For Manage Members

- **manage_members()**: This function provides a menu interface for managing members, including adding new members, viewing the member list, and checking membership expiration.

- **add_new_member():** This function handles the process of adding a new member or updating an existing member's membership.
  - If the member ID already exists in the "members.txt" file, it checks if the user wants to add a new membership for the existing member.
  - If yes, it calculates the new total membership duration, updates the member data in the "members.txt" file, and creates a new invoice using the create_invoice() function.

If the member ID doesn't exist, it prompts the user to enter the member's name, ID, and

  - phone number (with validation for a valid 10-digit phone number format).
  - It then prompts the user to select the membership duration and calculates the corresponding membership fee.
  - Finally, it saves the new member data to the "members.txt" file using the save_data_to_file() function and creates a new invoice using the create_invoice() function.

- **view_member_list():** This function reads the member data from the "members.txt" file using the read_data_from_file() function and **displays** the member's name, ID, phone, and membership duration.

- **check_membership_expiration_by_id():** This function prompts the user to enter a member ID, then retrieves the member's data from the "members.txt" file. It calculates the membership expiration date based on the membership duration and the invoice date, and displays the member's name, phone, expiration date, and the number of days left before expiration. If the membership is about to expire within 7 days, it also sends a notification to the member.

For Manage trainer

- **manage_trainers()**
  - - This function provides a menu interface for managing trainers.
  - - It displays a menu with three options: "Add New trainer", "View trainers List", and "Exit".
  - - The user can select an option by entering the corresponding number (1, 2, or 3).
  - - Depending on the user's choice, the function calls the `add_new_trainer()` or `view_trainer_list()` function or exits the loop.

- **add_new_trainer():** This function prompts the user to enter a trainer's name, email, and phone number (**with validation for a valid 10-digit phone number format**). It then saves the trainer's data to the "trainers.txt" file using the save_data_to_file() function.
  - If the phone number is valid, the function saves the trainer data (name, email, and phone number) to the "trainers.txt" file using the save_data_to_file() function.
  - If the phone number is invalid, the function prompts the user to enter a valid phone number.

- **view_trainer_list()**
  - This function is responsible for displaying the list of trainers.
  - It reads the trainer data from the "trainers.txt" file using the read_data_from_file() function.
  - If there is trainer data, the function splits the data by the comma (`,`) delimiter and prints the trainer information (name, email, and phone number) in a formatted manner.
  - If there is no trainer data, the function displays a message indicating that no trainers have been found.

For validation (date, time and day)

- **validate_date()**
  - This function is responsible for validating a date string.
  - It takes a date string as input and checks if it matches the expected format (YYYY-MM-DD) using a regular expression.
  - If the date string is valid, the function returns 0 (success), otherwise, it returns 1 (failure).

- **validate_time()**
  - This function is responsible for validating a time string.
  - It takes a time string as input and checks if it matches the expected format (HH:MM) using a regular expression.
  - If the time string is valid, the function returns 0 (success), otherwise, it returns 1 (failure).

- **validate_day()**
  - This function is responsible for validating a day string.
  - It takes a day string as input and converts it to lowercase.
  - It then checks if the day string matches any of the weekdays (Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday).
  - If the day string is valid, the function returns 0 (success), otherwise, it returns 1 (failure).

For Manage classes

- **manage_classes()**

  a menu-based interface for managing classes.
  - It allows the user to choose between adding a new class or viewing the class schedule.
  - It calls the add_class ()and view_class_schedule ()functions based on the user's choice.

- **add_class()**
  - responsible for adding a new class to the class schedule.
  - It prompts the user to enter the class name, the day of the week, and the time of the class.
  - It validates the user input for the day and time using the validate_day() and validate_time()
  - It displays the list of trainers from the" trainers.txt "file and prompts the user to enter the trainer name.
  - It checks if the entered trainer name exists in the" trainers.txt "file using the `grep` command.
  - If the trainer is found, it appends the class details) class name, day, time, and trainer name (to the" schedule.txt "file and displays a success message.
  - If the trainer is not found, it displays an error message.

- **view_class_schedule ()**
  - is responsible for displaying the current class schedule.
  - It reads the contents of the" schedule.txt "file and sorts the schedule by day and time using the `sort` command.
  - It then prints the class details) class name, time, and trainer name (for each class, grouping the classes by day.
  - If there are no classes in the schedule, it displays a message indicating that.

The main

- **main_menu ()**

is the entry point of the Gym Management System.
  - It displays the main menu options for managing members, trainers, and classes.
  - It uses a `case` statement to handle the user's choice and call the appropriate function manage_members(),manage_trainers(), or manage_classes()
  - If the user chooses to exit, the function terminates the program.

Using GitHub(ghala)

```
ghala@ghala-VirtualBox:~$ mkdir gym_project
ghala@ghala-VirtualBox:~$ cd gym_project
ghala@ghala-VirtualBox:~/gym_project$ git clone https://github.com/GhalaMansour/gym_system
.git
Cloning into 'gym_system'...
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 1), reused 5 (delta 0), pack-reused 0
Receiving objects: 100% (10/10), 5.53 KiB | 5.53 MiB/s, done.
Resolving deltas: 100% (1/1), done.
ghala@ghala-VirtualBox:~/gym_project$ ls
gym_system
ghala@ghala-VirtualBox:~/gym_project$ cd gym_system
ghala@ghala-VirtualBox:~/gym_project/gym_system$ ls
ghala@ghala-VirtualBox:~/gym_project/gym_system$ touch gym_management_system.sh
ghala@ghala-VirtualBox:~/gym_project/gym_system$ ls
gym_management_system.sh
ghala@ghala-VirtualBox:~/gym_project/gym_system$ cat > gym_management_system.sh
#!/bin/bash

# Helper functions for file-based storage
save_data_to_file() {
    local filename="$1"
    local data="$2"

    if [ -f "$filename" ]; then
        # Append the new data to the existing file
        echo "$data" >> "$filename"
    else
        # Create a new file with the data
```

```
        # Create a new file with the data
        echo "$data" > "$filename"
    fi
}

read_data_from_file() {
    local file_path="$1"

    if [ -f "$file_path" ]; then
        cat "$file_path"
    else
        echo ""
    fi
}

# Function to create an invoice
create_invoice() {
    local duration="$1"
    local fee="$2"
    local invoice_number=$((RANDOM % 1000 + 1))
    local invoice_date=$(date +"%Y-%m-%d")

    local invoice_data="
-----------------------------------------------------
Member Name: $member_name
Member ID: $member_id
Member Phone: \"$formatted_phone\"
Membership Duration: $duration months
Membership Fee: $fee
Invoice Number: $invoice_number
Invoice Date: $invoice_date
-----------------------------------------------------
"
```

```
"
    save_data_to_file "invoice_data.txt" "$invoice_data"
}


# Function to add a new member
add_new_member() {
    read -p "Enter member name: " member_name
    read -p "Enter member ID: " member_id

    # Check if the member ID already exists
    member_data=$(grep "$member_id" "members.txt")
    if [ -n "$member_data" ]; then
        echo "Member with ID $member_id already exists in the system."
        read -p "Do you want to add a new membership for this member? (y/n) " add_membersh
ip
        if [ "$add_membership" = "y" ]; then
            # Existing member, add new membership
            read -p "Select new membership duration (3, 6, 9, or 12 months): " new_members
hip_duration
            case $new_membership_duration in
                3)
                    new_membership_fee=99.99
                    ;;
                6)
                    new_membership_fee=179.99
                    ;;
                9)
                    new_membership_fee=249.99
                    ;;
                12)
                    new_membership_fee=299.99
                    ;;
```

```
                    ;;
                12)
                    new_membership_fee=299.99
                    ;;
                *)
                    echo "Invalid membership duration. Please try again."
                    add_new_member
                    return
                    ;;
            esac

            # Get the existing membership details from the member data
            IFS=', ' read -ra member_info <<< "$member_data"
            existing_membership_duration="${member_info[3]}"
            #existing_membership_fee="${member_info[4]}"

            # Calculate the new total membership duration
            if [[ "$existing_membership_duration" =~ ^[0-9]+$ ]]; then
                new_total_duration=$((existing_membership_duration + new_membership_durati
on))
            else
                new_total_duration=$new_membership_duration
            fi
create_invoice "$new_membership_duration" "$new_membership_fee"

            # Update the existing member data in members.txt
            new_member_data="$member_name, $member_id, \"$formatted_phone\", $new_total_du
ration"
            sed -i "/$member_id/d" "members.txt"
            save_data_to_file "members.txt" "$new_member_data"
        else
            echo "Okay, not adding a new membership."
        fi
```

```
            echo "Okay, not adding a new membership."
        fi
    else
        # New member
        local phone_number_pattern='^[0-9]{10}$'
        while true; do
        read -p "Enter member phone number: " member_phone
        if [[ "$member_phone" =~ $phone_number_pattern ]]; then
            # Format the phone number
            formatted_phone="${member_phone:0:3}-${member_phone:3:3}-${member_phone:6}"
        read -p "Select membership duration (3, 6, 9, or 12 months): " membership_duration

        case $membership_duration in
            3)
                membership_fee=99.99
                ;;
            6)
                membership_fee=179.99
                ;;
            9)
                membership_fee=249.99
                ;;
            12)
                membership_fee=299.99
                ;;
            *)
                echo "Invalid membership duration. Please try again."
                add_new_member
                return
                ;;
        esac
create_invoice "$membership_duration" "$membership_fee"
        # Save new member data to file
```

```
                ;;
        esac
create_invoice "$membership_duration" "$membership_fee"
        # Save new member data to file
        local member_data="$member_name, $member_id, \"$formatted_phone\", $membership_dur
ation"
        save_data_to_file "members.txt" "$member_data"
        break
        else
        echo "Invalid phone number format. Please enter a 10-digit phone number without an
y dashes or spaces."
        fi
        done
    fi
}


# Function to view member list
view_member_list() {
    local member_data=$(read_data_from_file "members.txt")
    if [ -n "$member_data" ]; then
        IFS=$','
        while read -ra member_info; do
            echo "name:${member_info[0]} ID:${member_info[1]} Phone:${member_info[2]} Memb
ership duration:${member_info[3]}
------------------------------------------------------------- "
        done <<< "$member_data"
    else
        echo "No members found."
    fi
}
```

```
# Function to check membership expiration by member ID
check_membership_expiration_by_id() {
    read -p "Enter member ID: " member_id
    local member_data=$(grep "$member_id" "members.txt")
    if [ -n "$member_data" ]; then
        IFS=$','
        read -ra member_info <<< "$member_data"
        local member_name="${member_info[0]}"
        local member_phone="${member_info[2]}"
        local membership_duration="${member_info[3]}"
        local invoice_date="${member_info[6]}"

        # Calculate expiration date
        local expiration_date=$(date -d "$invoice_date + $membership_duration months" +"%Y
-%m-%d")
        local days_left=$(($(date -d "$expiration_date" +"%s") - $(date +"%s")))
        days_left=$((days_left / 86400))

        echo "Member: $member_name, Phone:\"$member_phone\", Expiration: $expiration_date,
 Days Left: $days_left"

        if [ "$days_left" -le 7 ]; then
            echo "
This is to inform you that the GYJ Gym membership for "$member_name" (Phone: \"$formatted_
phone\") will expire in $days_left days. Please make sure to follow up with them regarding
 membership renewal."

        fi
    else
        echo "Member with ID $member_id not found."
    fi
}
```

```
manage_members(){
while true; do
        echo "======= Members Management ========"
        echo "1. Add New Member"
        echo "2. View Member List"
        echo "3. Check Membership Expiration"
        echo "4. Exit"

        read -p "Enter your choice (1-4): " choice

        case $choice in
            1)
                add_new_member
                ;;
            2)
                view_member_list
                ;;
            3)
                check_membership_expiration_by_id
                ;;

            4)
                echo "Back to the main menu..."
                break
                ;;
            *)
                echo "Invalid choice. Please try again."
                ;;
        esac
    done

}
```

```
  (use "git add <file>..." to include in what will be committed)
        gym_management_system.sh

nothing added to commit but untracked files present (use "git add" to track)
ghala@ghala-VirtualBox:~/gym_project/gym_system$ git add gym_management_system.sh
ghala@ghala-VirtualBox:~/gym_project/gym_system$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   gym_management_system.sh

ghala@ghala-VirtualBox:~/gym_project/gym_system$ git commit -m "ghala (members management)
 commit"
[main 7fda645] ghala (members management) commit
 1 file changed, 219 insertions(+)
 create mode 100644 gym_management_system.sh
ghala@ghala-VirtualBox:~/gym_project/gym_system$ git remote set-url origin https://ghp_HCV
6PzIUxoSzuKrh8Zd2dh69RFTlvw0v7kfK@github.com/GhalaMansour/gym_system.git
ghala@ghala-VirtualBox:~/gym_project/gym_system$ git remote -v
origin  https://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@github.com/GhalaMansour/gym_syste
m.git (fetch)
origin  https://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@github.com/GhalaMansour/gym_syste
m.git (push)
ghala@ghala-VirtualBox:~/gym_project/gym_system$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 2.04 KiB | 2.04 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/GhalaMansour/gym_system.git
   1e60a8e..7fda645  main -> main
ghala@ghala-VirtualBox:~/gym_project/gym_system$ chmod +x gym_management_system.sh
```

Using GitHub(yara)

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ git pull origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 7 (delta 1), reused 2 (delta 0), pack-reused 0
Unpacking objects: 100% (7/7), 1.99 KiB | 145.00 KiB/s, done.
From https://github.com/GhalaMansour/gym_system
 * branch            main       -> FETCH_HEAD
   98b3589..7fda645  main       -> origin/main
Updating 98b3589..7fda645
Fast-forward
 gym_management_system.sh | 219 +---------------------------------------------
 manage_trainers.sh       |  62 --------------
 2 files changed, 1 insertion(+), 280 deletions(-)
 delete mode 100644 manage_trainers.sh
ubuntu@yaraalsiamy:~/workspace/gym_system$ ls
gym_management_system.sh
```

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ cat >> gym_management_system.sh


# Function to add a new trainer
add_new_trainer() {
    read -p "Enter trainer name: " trainer_name
    read -p "Enter trainer email: " trainer_email

    local phone_number_pattern='^[0-9]{10}$'
    while true; do
        read -p "Enter trainer phone number: " trainer_phone
        if [[ "$trainer_phone" =~ $phone_number_pattern ]]; then
            # Save trainer data to file
            local trainer_data="$trainer_name,$trainer_email,$trainer_phone"
            save_data_to_file "trainers.txt" "$trainer_data"
            echo "New trainer added successfully!"
            break
        else
            echo "Invalid phone number format. Please enter a 10-digit phone number without any dashes or spaces."
        fi
    done
}

# Function to view trainer list
view_trainer_list() {
    local trainer_data=$(read_data_from_file "trainers.txt")
    if [ -n "$trainer_data" ]; then
        IFS=','
        while read -ra trainer_info; do
            echo "Name: ${trainer_info[0]} Email: ${trainer_info[1]} Phone: ${trainer_info[2]}
-------------------------------------------------------------"
        done <<< "$trainer_data"
    else
        echo "No trainers found."
```

```
    else
        echo "No trainers found."
    fi
}

# Manage trainers menu
manage_trainers(){
while true; do
        echo "======= Trainers Management ======="
        echo "1. Add New trainer"
        echo "2. View trainers List"
        echo "3. Exit"

        read -p "Enter your choice (1-3): " choice

        case $choice in
            1)
                add_new_trainer
                ;;
            2)
                view_trainer_list
                ;;
            3)
                echo "Back to the main menu..."
                break
                ;;
            *)
                echo "Invalid choice. Please try again."
                ;;
        esac
    done
}
^C
```

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ git add gym_management_system.sh
ubuntu@yaraalsiamy:~/workspace/gym_system$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   gym_management_system.sh

ubuntu@yaraalsiamy:~/workspace/gym_system$ git commit -m "trainers management"
[main 0778728] trainers management
 1 file changed, 63 insertions(+)
ubuntu@yaraalsiamy:~/workspace/gym_system$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 5 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 773 bytes | 257.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/GhalaMansour/gym_system.git
   7fda645..0778728  main -> main
ubuntu@yaraalsiamy:~/workspace/gym_system$
```

Using GitHub(Joud)

Joud use Yara's device

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ cat >> gym_management_system.sh


validate_date() {
    local date_str=$1
    local date_regex='^[0-9]{4}-[0-9]{2}-[0-9]{2}$'

    if [[ $date_str =~ $date_regex ]]; then
        return 0
    else
        return 1
    fi
}

validate_time() {
    local time_str=$1
    local time_regex='^[0-9]{2}\:[0-9]{2}$'

    if [[ $time_str =~ $time_regex ]]; then
        return 0
    else
        return 1
    fi
}
validate_day() {
    local day_str=$(echo "$1" | tr '[:upper:]' '[:lower:]')
    local days=("monday" "tuesday" "wednesday" "thursday" "friday" "saturday" "sunday")

    for day in "${days[@]}"; do
        if [[ "$day_str" == "$day" ]]; then
            return 0
        fi
    done
```

```
    done

    return 1
}

add_class() {
    read -p "Enter the class name: " class_name
    while true; do
        read -p "Enter the day of the week (e.g., Monday, Tuesday, etc.): " class_day
        if validate_day "$class_day"; then
            break
        else
            echo "Invalid day format. Please enter a valid day of the week."
        fi
    done

    while true; do
        read -p "Enter the time (HH:MM): " class_time
        if validate_time "$class_time"; then
            break
        else
            echo "Invalid time format. Please use the format HH:MM."
        fi
    done

    # Display the list of trainers
    echo "Select a trainer:"
    while IFS="," read -r name email phone; do
        echo "- $name"
    done < "trainers.txt"

    read -p "Enter the trainer name: " trainer_name

    # Check if the trainer exists
```

```
    # Check if the trainer exists
    if grep -q "$trainer_name" "trainers.txt"; then
        # Append class details to the schedule file
        echo "$class_name, $class_day, $class_time, $trainer_name" >> "schedule.txt"
        echo "Class added to the schedule successfully!"
    else
        echo "Trainer not found. Please add the trainer first."
    fi
}

view_class_schedule() {
    local schedule_data=$(cat "schedule.txt")
    if [ -n "$schedule_data" ]; then
        # Sort the schedule by day and time
        sorted_schedule=$(echo "$schedule_data" | sort -k2,2 -k3,3)

        current_day=""
        while IFS=',' read -r class_name class_day class_time trainer_name; do
            if [[ "$class_day" != "$current_day" ]]; then
                echo "====== $class_day ========"
                current_day="$class_day"
            fi
            echo "Class: $class_name"
            echo "Time: $class_time"
            echo "Trainer: $trainer_name"
            echo "---"
        done <<< "$sorted_schedule"
    else
        echo "No classes found in the schedule."
    fi
}

# Manage classes menu
manage_classes(){
```

```
# Manage classes menu
manage_classes(){
while true; do
        echo "======= Classes Management ======="
        echo "1. Add New class"
        echo "2. View class schedule"
        echo "3. Exit"

        read -p "Enter your choice (1-3): " choice

        case $choice in
            1)
                add_class
                ;;
            2)
                view_class_schedule
                ;;
            3)
                echo "Back to the main menu..."
                break
                ;;
            *)
                echo "Invalid choice. Please try again."
                ;;
        esac
    done
}
^C
ubuntu@yaraalsiamy:~/workspace/gym_system$ git add gym_management_system.sh
ubuntu@yaraalsiamy:~/workspace/gym_system$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
```

```
            3)
                echo "Back to the main menu..."
                break
                ;;
            *)
                echo "Invalid choice. Please try again."
                ;;
        esac
    done
}
^C
ubuntu@yaraalsiamy:~/workspace/gym_system$ git add gym_management_system.sh
ubuntu@yaraalsiamy:~/workspace/gym_system$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   gym_management_system.sh

ubuntu@yaraalsiamy:~/workspace/gym_system$ git commit -m "classes management"
[main 752fe34] classes management
 1 file changed, 123 insertions(+)
ubuntu@yaraalsiamy:~/workspace/gym_system$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 5 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.22 KiB | 313.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/GhalaMansour/gym_system.git
   0778728..752fe34  main -> main
ubuntu@yaraalsiamy:~/workspace/gym_system$
```

Using GitHub(main menu)

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ git add gym_management_system.sh
ubuntu@yaraalsiamy:~/workspace/gym_system$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   gym_management_system.sh

ubuntu@yaraalsiamy:~/workspace/gym_system$ git commit -m "Main menu"
[main 8c1edb5] Main menu
 1 file changed, 31 insertions(+)
ubuntu@yaraalsiamy:~/workspace/gym_system$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 5 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/GhalaMansour/gym_system.git
   752fe34..8c1edb5  main -> main
ubuntu@yaraalsiamy:~/workspace/gym_system$
```

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ cat >> gym_management_system.sh

main_menu() {
    while true; do
        echo "===== Gym Management System ======"
        echo "1. Members Management"
        echo "2. Trainers Management"
        echo "3. Classes Management"
        echo "4. Exit"

        read -p "Enter your choice (1-4): " choice

        case $choice in
            1)
                manage_members
                ;;
            2)
                manage_trainers
                ;;
            3)
                manage_classes
                ;;
            4)
                echo "Exiting Gym Management System..."
                break
                ;;
            *)
                echo "Invalid choice. Please try again."
                ;;
        esac
    done
}
^C
```

Output (Members management menu)

```
ghala@ghala-VirtualBox: ~/gym_project/gym_system
ghala@ghala-VirtualBox:~/gym_project/gym_system$ ./gym_management_system.sh
===== Gym Management System =====
1. Members Management
2. Trainers Management
3. Classes Management
4. Exit
Enter your choice (1-4): 1
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 1
Enter member name: Ghala
Enter member ID: 444007247
Enter member phone number: 0569777300
Select membership duration (3, 6, 9, or 12 months): 3
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 1
Enter member name: Ghala
Enter member ID: 444007247
Member with ID 444007247 already exists in the system.
Do you want to add a new membership for this member? (y/n) y
Select new membership duration (3, 6, 9, or 12 months): 9
======= Members Management =======
1. Add New Member
```

```
members.txt
1 Ghala, 444007247, "056-977-7300", 3
```

```
members.txt
1 Ghala, 444007247, "056-977-7300", 12
```

```
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 2
name:Ghala ID: 444007247 Phone: "056-977-7300" Membership duration: 12
-----------------------------------------------------------------
name:yara ID: 22 Phone: "056-652-4933" Membership duration: 6
-----------------------------------------------------------------
name:joud ID: 123 Phone: "057-353-8222" Membership duration: 9
-----------------------------------------------------------------
```

```
members.txt
1 Ghala, 444007247, "056-977-7300", 12
2 yara, 22, "056-652-4933", 6
3 joud, 123, "057-353-8222", 9
```

```
invoice_data.txt
1
2 ----------------------------------------------------
3 Member Name: Ghala
4 Member ID: 444007247
5 Member Phone: "056-977-7300"
6 Membership Duration: 3 months
7 Membership Fee: 99.99
8 Invoice Number: 237
9 Invoice Date: 2024-05-22
10 ----------------------------------------------------
11
12
13 ----------------------------------------------------
14 Member Name: Ghala
15 Member ID: 444007247
16 Member Phone: "056-977-7300"
17 Membership Duration: 9 months
18 Membership Fee: 249.99
19 Invoice Number: 749
20 Invoice Date: 2024-05-22
21 ----------------------------------------------------
22
23
24 ----------------------------------------------------
25 Member Name: yara
26 Member ID: 22
27 Member Phone: "056-652-4933"
28 Membership Duration: 6 months
29 Membership Fee: 179.99
30 Invoice Number: 683
31 Invoice Date: 2024-05-22
32 ----------------------------------------------------
33
34
35 ----------------------------------------------------
36 Member Name: joud
37 Member ID: 123
38 Member Phone: "057-353-8222"
39 Membership Duration: 9 months
40 Membership Fee: 249.99
41 Invoice Number: 742
42 Invoice Date: 2024-05-22
43 ----------------------------------------------------
```

```
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 3
Enter member ID: 444007247
Member: Ghala, Phone:" "056-977-7300"", Expiration: 2025-05-22, Days Left: 364
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 3
Enter member ID: 123
Member: joud, Phone:" "057-353-8222"", Expiration: 2025-02-22, Days Left: 275
======= Members Management =======
1. Add New Member
2. View Member List
3. Check Membership Expiration
4. Exit
Enter your choice (1-4): 4
Back to the main menu...
===== Gym Management System =====
```

Output (Trainers management menu)

```
ubuntu@yaraalsiamy:~/workspace/gym_system$ ./gym_management_system.sh
===== Gym Management System =====
1. Members Management
2. Trainers Management
3. Classes Management
4. Exit
Enter your choice (1-4): 2
======= Trainers Management =======
1. Add New trainer
2. View trainers List
3. Exit
Enter your choice (1-3): 1
Enter trainer name: leen
Enter trainer email: leen@gmail.com
Enter trainer phone number: 0554398765
New trainer added successfully!
======= Trainers Management =======
1. Add New trainer
2. View trainers List
3. Exit
Enter your choice (1-3): 2
Name: leen Email: leen@gmail.com Phone: 0554398765
-----------------------------------------------------------
======= Trainers Management =======
1. Add New trainer
2. View trainers List
3. Exit
Enter your choice (1-3):
```

```
leen,leen@gmail.com,0554398765
```

Trainers.txt

Output (classes management menu)

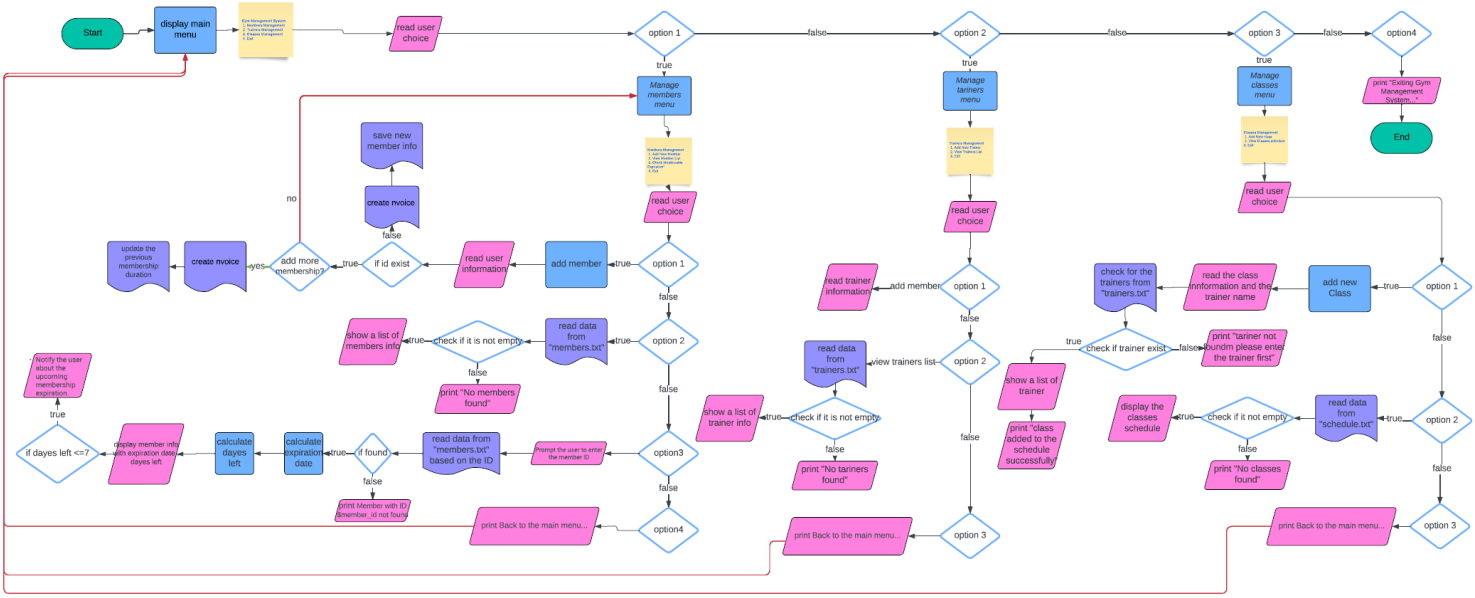
```
======= Classes Management =======
1. Add New class
2. View class schedule
3. Exit
Enter your choice (1-3): 1
Enter the class name: yoga
Enter the day of the week (e.g., Monday, Tuesday, etc.): sunday
Enter the time (HH:MM): 08:30
Select a trainer:
- leen
Enter the trainer name: leen
Class added to the schedule successfully!
======= Classes Management =======
1. Add New class
2. View class schedule
3. Exit
Enter your choice (1-3): 2
======  sunday =======
Class: yoga
Time:  08:30
Trainer:  leen
---
======= Classes Management =======
1. Add New class
2. View class schedule
3. Exit
Enter your choice (1-3):
```

```
yoga, sunday, 08:30, leen
```

Schedule.txt

Flowchart

flowchart link:https://lucid.app/lucidchart/22c7e514-d634-4f15-9238-4cf978e22814/edit?viewport_loc=274%2C39%2C1886%2C2549%2C0_0&invitationId=inv_a6dbe547-04a2-48dd-ba05-632214af65be



About out work

Joud	**Coding** for classes management & **flowchart**
Ghala	**Coding** for members management & **flowchart**
Yara	**Coding** for trainer management & **flowchart**
All of us	Writing the **report** & the **presentation** We all used **GitHub** to complete this project and share it together.

Conclusion:

The Gym Management System simplifies the process of managing a gym facility by automating key tasks such as member management, membership expiration tracking, trainer management, and class scheduling. This software enhances efficiency, improves organization, and enables effective administration of gym operations.